

Project:-Netflix Customer churn analysis

Tool Used:

- Excel
- SQL Server
- Power BI

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Executive Summary

Project Overview

The streaming industry has become increasingly competitive, making customer retention a critical success factor for subscription-based platforms such as Netflix. Retaining existing customers is often more cost-effective than acquiring new ones, making it essential for businesses to understand the factors that influence customer churn.

This project focuses on analyzing customer subscription and engagement data to identify patterns associated with customer attrition. The analysis examines various factors such as subscription plans, customer demographics, watch behavior, satisfaction scores, payment failures, and account activity. By leveraging data analytics techniques, the project aims to uncover the key drivers behind customer churn and evaluate their impact on business performance.

The ultimate objective of this analysis is to help Netflix improve customer retention, enhance user satisfaction, reduce revenue leakage, and develop targeted strategies that strengthen long-term customer relationships.

Business Problem

Customer churn is one of the most significant challenges faced by subscription-based businesses. When customers cancel their subscriptions, organizations not only lose recurring revenue but also incur additional costs associated with acquiring new customers. High churn rates can negatively impact profitability, customer lifetime value, and overall business growth.

This project addresses the challenge by analyzing customer data to answer critical business questions such as:

- Which subscription plans experience the highest churn?
- Which customer segments are most likely to discontinue their subscriptions?
- How do satisfaction scores influence customer retention?
- What role do payment failures play in customer attrition?
- Which countries contribute the most to churn-related revenue loss?

Dataset & Methodology

Dataset Information

Mention:

- 25,000 customer records
- Customer demographics
- Subscription details
- Watch behavior
- Satisfaction scores
- Payment history
- Churn status

Data Preparation:

- Data Cleaning in Excel
- Age Category Creation
- Missing Value Handling
- Data Validation

Tools Used

Tool	Purpose
Excel	Data Cleaning
SQL Server	Data Analysis
Power BI	Dashboard Development

SQL Analysis & Findings:

To understand customer churn behavior and identify the key factors influencing customer retention, multiple SQL queries were performed on the customer dataset. The analysis focused on churn metrics, customer segmentation, subscription plans, revenue impact, satisfaction levels, payment behavior, and regional trends.

Key Metrics:

Metric	Value
Total Customers	25,000
Churned Customers	14,258
Churn Rate	57.03%
Revenue Lost	6.18M

The overall churn rate of **57.03%** indicates that more than half of the customers in the dataset discontinued their subscriptions. This resulted in an estimated revenue loss of **6.18M**, highlighting the significant financial impact of customer attrition on the business.

Key SQL Insights:

Premium plan showed the highest churn volume.

Adult customers experienced higher churn compared to other age groups.

Customers with low satisfaction scores were more likely to churn.

Frequent payment failures increased churn risk.

Dashboard Overview

An interactive Power BI dashboard was developed to provide a comprehensive view of customer churn behavior, customer engagement, and business performance metrics. The dashboard enables stakeholders to monitor key churn indicators, identify high-risk customer segments, and evaluate the impact of customer attrition on revenue.

The dashboard combines multiple visualizations to simplify data interpretation and support data-driven decision-making.

Customer Churn Distribution

Displays the proportion of active and churned customers, providing a high-level overview of customer retention performance.

Subscription Plan Wise Churn

Analyzes churn across different subscription plans to identify which plans experience the highest customer loss and require retention-focused strategies.

Country Wise Churn

Highlights countries contributing the highest number of churned customers, helping identify regional retention challenges and market-specific trends.

Satisfaction Score vs Churn

Evaluates the relationship between customer satisfaction and churn behavior, enabling the identification of dissatisfaction-driven customer attrition.

Average Watch Hours by Favorite Genre

Measures customer engagement across content genres and identifies the most preferred content categories based on average watch time.

Active Customers by Country

Displays the distribution of active customers across different countries, helping identify strong-performing markets and customer retention opportunities.

KPI Cards

The dashboard includes key performance indicators such as Total Customers, Active Customers, Churned Customers, Churn Rate, and Revenue Lost, providing stakeholders with a quick summary of overall business performance.

Key Business Insights

Key Findings:

- 1]-Overall churn rate reached 57.03%.
- 2]-Premium subscribers recorded the highest churn count.
- 3]-Low satisfaction scores strongly influenced churn behavior.
- 4]-Payment failures were a major contributor to customer attrition.
- 5]-Australia showed the highest churn volume.
- 6]-Sci-Fi viewers recorded the highest average watch hours.
- 7]-Customer churn resulted in an estimated revenue loss of 6.18M.

Recommendations & Conclusion

Recommendations:

- Improve customer satisfaction through personalized content recommendations.
- Reduce payment failures using automated billing reminders.
- Target high-risk customers with retention campaigns.
- Focus retention efforts on high-revenue subscription plans.
- Develop region-specific engagement strategies in high-churn countries.

Conclusion

The analysis identified several key factors contributing to customer churn, including low satisfaction scores, payment failures, and subscription plan behavior. By addressing these issues, Netflix can improve customer retention, reduce revenue loss, and enhance overall customer experience.